

Proposed additions to the Phase 1 Concept Proposal

Team AuraQu, allosteric branch, 2026-09-12. Nothing here is in the submission yet. Each block is staged with **status:** PENDING; the merge tool writes a new file and never edits the proposal in place. The base is PHASE1_SUBMISSION_V4.md, byte-identical to bartosz@9f1d521.

Consistency was checked mechanically against the 64 numeric claims already in the proposal. Where a number differs from one already there, it is a second measurement on a second cohort and says so in the same sentence.

ADD-1 — Protein-identity floor, second cohort

To: 5. Validation Plan. *Cost:* 2 lines. *Status:* PENDING.

Inserted after: “...re without this floor is not interpretable, and we have found no report of the check.”

Re-measured on the second cohort described in Appendix B, the same constant-per-protein score reaches **AUC 0.771** across 630 proteins — above every method either track has tested, and a wider margin than the 0.65 above.

ADD-2 — Independent replication of the central null

To: 2. Technical Approach. *Cost:* 4 lines. *Status:* PENDING.

Inserted after: “...because McNemar is a paired test over the same families.”

A second pipeline, built independently on a different detector and a different candidate filter, reaches the same verdict on a larger cohort: across 630 proteins in 399 families, the seeded walk loses to closeness centrality at every distal filter setting (family-weighted AUC 0.585 / 0.542 / 0.544 / 0.543 against 0.601 / 0.569 / 0.575 / 0.580; Wilcoxon $p = 0.001, 0.003, 0.044, 0.258$). The two cohorts are not pooled — they differ in detector and in family convention — so this is replication, not an enlarged n . Appendix B reports it in full.

ADD-3 — A blind model over all 221 observables reproduces distance

To: 2. Technical Approach. *Cost:* 3 lines. *Status:* PENDING.

Inserted after: “...What it adds is simply not coherence”
The same conclusion survives combining the observ-

ables rather than choosing among them. A logistic regression over all 221 per-residue score vectors, true leave-one-family-out, reaches family-weighted AUC **0.7216** on the distal subset — against 0.5405 for the best single cell selected on the same data, and **0.7219** for the trivial reversed-distance floor. The difference from the floor is -0.0003 (Wilcoxon $p = 0.90$): optimally combined, the quantum observables reproduce distance and nothing further.

ADD-4 — Appendix B, the full second-cohort report

To: Appendix (new B, after references). *Cost:* ~1 page. *Status:* PENDING.

(The full appendix drafted earlier — pipeline description, the four-MIN_HOP table, the LOFO result, the two-source interference run, the family-provenance audit, the notebook run, and our retractions. Held until ADD-1..3 are settled, since it must not repeat whatever goes into the body.)

ADD-5 — Our retractions in the §7 ledger

To: 7. Team Capability. *Cost:* 2 lines. *Status:* PENDING.

Inserted after: “...It found five of our own errors in seven days, including our headline result.”

The same register operating on the second branch caught three more: an AUC read backwards in a distal proximity-floor claim, a classical-comparison arm that differed in operator as well as in coherence, and a blind positive filed as a to-do rather than reported. All three are corrected in the public history.

ADD-6 — The thirteen operators and seventeen scores, named

To: 2. Technical Approach (Paradigm). *Cost:* ~16 lines. *Status:* PENDING.

Inserted after: “...-body objects and conformational ensembles rather than single-structure optimisation.”

What was swept. Residues are nodes at their C-alpha coordinates, edges within a 10 Å cutoff; the walk is $U(t) = \exp(-iHt)$ seeded at the active site, reported in the converged time-average $p_{avg}(s \rightarrow a) = \sum_k |v_k(s)|^2 |v_k(a)|^2$ — a sum of squares, which is why the converged propagator is phase-free. Every operator is real symmetric, hence Hermitian.

Thirteen operators = four edge weightings x three normalisations, plus the challenge-specified H_{new} . The weightings are binary ($d < 10$ Å), exponential $\exp(-0.3 d)$, Gaussian $\exp(-d^2/72)$ and harmonic $1/(d + 0.5)^2$; each is used under three normalisa-

block	to	lines
ADD-1	S5	2
ADD-2	S2	4
ADD-3	S2	3
ADD-4	App B	1 page
ADD-5	S7	2
ADD-6	S2	16
ADD-7	S1 / App B	10

tions — the adjacency W itself, the combinatorial Laplacian $L = D - W$, and the symmetric normalised $L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}$ — giving twelve, written `binary/adj` through `harm/sym`.

H_{new} is the thirteenth: L_{sym} of the exponential graph plus a diagonal site potential $0.08 V_B + 0.16 V_T + 0.08 V_R + 0.04 V_C + 0.04 V_M$, each term z-scored and signed so penalties raise the potential and rewards lower it — V_B B-factor disorder, V_T chain termini and solvent exposure, V_R rigidity (degree, clustering, inverse GNM fluctuation), V_C GNM covariance with the core, V_M participation in the ten slowest GNM modes. Weights pre-registered in the notebook, not fitted here.

Seventeen scores, in five families:

Thirteen times seventeen is the 221 cells per protein quoted above. One qualification belongs with this table rather than buried: `neg_dE`, the strongest single score, is time-independent by construction, and equals $\sqrt{\sum_j W_{ij}^2}$ to machine precision — a classical local statistic wearing a quantum name. It ranks well and it cannot carry interference, which is one measured reason the coherent and decoherent arms agree.

ADD-7 — KRAS G12C on 4LDJ: the seeding stage decides the target

To: 1. Problem Framing (beside the hit-list table) or Appendix B. *Cost:* ~10 lines. *Status:* PENDING.

The stage that decided KRAS was not the walk.

Re-running the mandated target end to end on 4LDJ — the genuine G12C apo structure (residue 12 = CYS; GDP and Mg only; we verified that six apo G12C depositions exist and that Table 1's 40BE is wild-type at residue 12 while the organisers' suggested 8S8C is drug-bound) — the result is decided before the quantum walk runs.

Candidate pockets come from fpocket and are cut to a top-K before seeding. Under the narrower setting (fpocket ranking, top-5 by druggability) the true sotora-sib pocket ranks **#6** and is discarded: 1 drug residue survives among 36 seeds, and P@5 is zero by construction — the pipeline cannot answer correctly whatever the operator does. Widening to a druggability-and-

allostery consensus at top-10 seeds it: **8 drug residues among 44**, base rate 0.182. Same structure, same walk, same scoring; the selection stage alone moved the target from unanswerable to answerable.

Scored on that seed set, the best of 14 operators x 17 scores reaches AUC **0.809** (H_{new} with the resolvent score at the band top, P@5 0.6) and the worst reaches **0.184** (`H11_aniso`, proximity-corrected residual) — a spread of 0.63 AUC across cells that differ only in operator and score. Read correctly the worst is not a failure but a reversal: $1 - 0.184 = 0.816$, within noise of the best. Under the pre-registered single cell fixed before scoring, the target returns an honest negative (label-permutation null, $p > 0.05$).

Consequence, and it is the same one section 6 reaches from the funnel ablation: on this target the classical selection stage, not the transport subroutine, determines whether the answer is reachable at all. A method reported only at its best cell, on a seed set chosen after the fact, would have reported 0.809 here.

family	scores	what it ranks a residue by
occupation	p_avg, p_peak, R	converged transfer, its maximum over the time grid, and their ratio
proximity-corrected	residLOG, residRAW, residLOG_dX, residRAW_dX, pavg_over_dX	transfer with distance and degree regressed out, raw or log, optionally focused by walker spread
resolvent	green_zero_0.01, green_zero_0.05, green_lmax_0.05	$ \langle \mathbf{a} (E + i \eta - H)^{-1} \mathbf{s} \rangle ^2$ at the band edge and band top — a commute-time / communicability analogue
dispersion	neg_dD_mean, neg_ED_final, QMI	spread of the walker's distance distribution, its final energy dispersion, and a mutual-information proxy
energy-uncertainty	neg_dE, residLOG_dE, pavg_over_dE	the seed state's spectral variance $\sqrt{\langle E^2 \rangle - \langle E \rangle^2}$, alone or combining with the above